

Mohamed H. Mousa, PhD

Computational Neuroscientist

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SUMMARY

Neuroengineer with 8+ years of experience in computational neuroscience, electrophysiology, neural signal processing, and biophysical modeling. Developed NEURON-based models of spinal circuits and electrical stimulation effects, built scalable Python/MATLAB/SpikesInterface pipelines for neural data analysis, and conducted human and animal electrophysiology experiments. Experience spanning neural decoding, EMG analysis, nonlinear neural dynamics, and machine learning for neural systems. Most recent work published as first author in the *Journal of Neuroscience* (2026).

EXPERIENCE

Temple University

Oct 2024 – Present

Postdoctoral Fellow Research Associate Philadelphia, PA

- Built end-to-end pipelines to process, decompose, and analyze electrophysiological data (128 channel) using MATLAB, Python, R, and SpikesInterface.
- Developed Python-based analysis pipelines to quantify functional connectivity and nonlinear dynamics in spinal interneuron networks using graph-theoretic and statistical methods.
- Conducted high-density electrode recordings from human subjects and animal models across multiple experimental conditions.

Wright State University

Jan 2023 – Oct 2024

Postdoctoral Researcher Dayton, OH

- Led IRB-approved clinical study to develop a diagnostic biomarker for ALS — enrolled, consented, and performed EMG recordings on human participants (20+).
- Analyzed and reverse-engineered electrophysiological recordings from human and animal models to identify disease signatures.
- Developed predictive biophysical computational models using NEURON and MATLAB to classify ALS-related neural changes.

Wright State University

Jun 2017 – Dec 2022

Doctoral Candidate Dayton, OH

- Developed 3D models of motoneurons to reverse-engineer experimental electrophysiological data.
- Modeled electrical field potentials and the effects of neural stimulation on motoneuron excitability.
- Automated analysis and feature extraction of electrophysiological datasets using Python and MATLAB, reducing analysis time by 60×.

Cairo University

Dec 2014 – Jun 2017

Research Assistant Cairo, Egypt

- Taught courses in medical equipment, bio-electronics, data structures, and computer vision.
- Supervised BCI graduation projects using P300 brain waves.

Maadi Military Hospital

2013 – 2014

Field Clinical Engineer Cairo, Egypt

- Supported installation, calibration, and maintenance of MRI, CT, and ventilator systems in clinical environments, coordinating directly with physicians and hospital engineering staff.

EDUCATION

- **PhD in Engineering — Computational Neuroscience**, Wright State University, Dayton, OH, 2022
Dissertation: Reverse Engineering Spinal Motoneuron Properties in Slice and Whole Cord Preparations Using Computational Modeling

SELECTED PUBLICATIONS

Full publication list available on [Google Scholar](#).

- **Mousa MH**, Zaback M, Lemay MA, Thompson CK (2026) Activity and Functional Connectivity of Lumbar Spinal Interneurons at Rest: Irregular, Sparse Yet Structured. *Journal of Neuroscience*, e0784262026.
- **Mousa MH**, Wages NP, Elbasiouny SM (2024) Onion skin is not a universal firing pattern for spinal motoneurons: simulation study. *Journal of Neurophysiology*, 132(1), 240–258.
- Gamal M, **Mousa MH**, Eldawlatly S, Elbasiouny SM (2021) In-silico development and assessment of a Kalman filter motor decoder for prosthetic hand control. *Comput Biol Med*, 132, 104353.
- Mahrous AA, **Mousa MH**, Elbasiouny SM (2019) The Mechanistic Basis for Successful Spinal Cord Stimulation to Generate Steady Motor Outputs. *Frontiers in Cellular Neuroscience*, 13, 359–374.

TECHNICAL SKILLS

Domain Expertise: Computational Neuroscience | Biophysical Modeling & Simulation | Signal & Image Processing | Network & Graph Theory | Statistical Analysis | Nonlinear Dynamics | Clinical Management & IRB Protocols

Programming Languages: Python | MATLAB | R | C/C++ | NEURON (Hoc, NMODL) | SQL

Libraries & Tools: NumPy | SciPy | Pandas | Matplotlib | Seaborn | Plotly | statsmodels | Pingouin | SpikeInterface | PyTorch (developing)

AWARDS

- SfN Trainee Professional Development Award, Society for Neuroscience, 2026
- Excellence award, top PhD graduate student in biomedical engineering, WSU, 2023
- Science Gateways Community Institute (SGCI) Science Ambassador — \$1,200 travel award, 2019
- Travel Award, Professional Development Grant, WSU, 2018
- Best paper award, Cairo International Biomedical Engineering (CIBEC), 2016
- Graduated First in Class with Distinction, Cairo University, 2013